

Kestrel Spectrograph: UV Channel Layout

1 Description

This note describes the optical layout of the Kestrel spectrograph's ultraviolet channel, from the beam it receives to its detector, and the guide camera that shares its input.

2 Arrangement

Figure 1 shows the arrangement. From the input beam, the light passes to the fold mirror FM1. From the fold mirror FM1, the light passes to the dichroic D1. From the dichroic D1, the light passes to OAP1. From OAP1, the light passes to the field stop. From the field stop, the light passes to the ellipsoidal mirror EM1. From the ellipsoidal mirror EM1, the light passes to the filter wheel. From the filter wheel, the light passes to the grism G1. From the grism G1, the light passes to EMCCD. From the dichroic D1, the light passes to the guide camera in guide mode. The field stop is located at a focal plane. The filter wheel is located at a pupil plane. The grism G1 is located at a pupil plane. The filter wheel is movable. The grism G1 is movable. OAP1 is part of UV channel. The field stop is part of UV channel. The ellipsoidal mirror EM1 is part of UV channel. The filter wheel is part of UV channel. The grism G1 is part of UV channel. EMCCD is part of UV channel. The field stop has two options: the wide stop and the narrow stop.

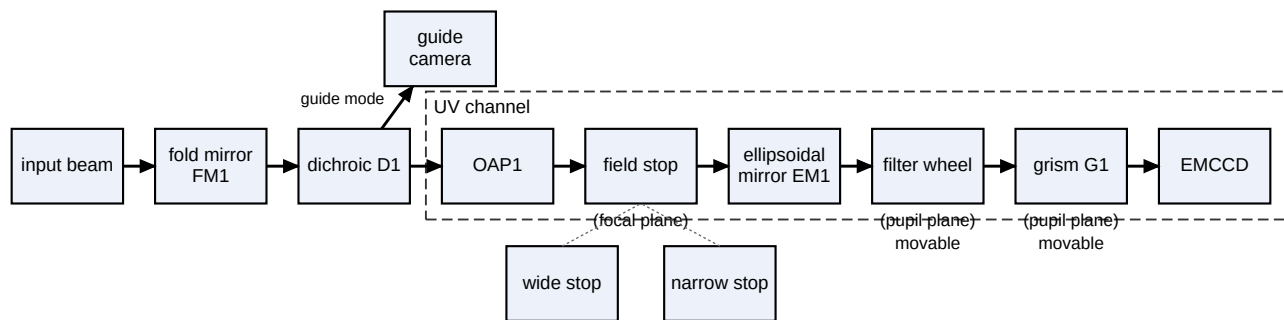


Figure 1. UV channel optical layout